UNIVERSITY COLLEGE DUBLIN

School of Chemistry

EXPERIMENT 1:
Determination of Quinine Concentration in Tonic Water Samples Using Fluorescence Spectroscopy

Module:	CHEM30110 - Instrumental Analysis
Student Name:	Anastasia Musetoiu
Date:	November 8, 2025
Experiment Type:	Quantitative Fluorescence Analysis

1. INTRODUCTION AND OBJECTIVES

Background

Quinine is an organic compound derived from the bark of the cinchona tree, historically used to prevent and treat malaria. Today, it remains a key ingredient in tonic water, where it provides the characteristic bitter taste. The European Union sets a safety limit of 100 mg/L (100 ppm) for quinine in tonic water and soft drinks to prevent potential adverse effects.

Quinine exhibits strong fluorescence properties, absorbing UV light at 350 nm and emitting in the visible spectrum at approximately 450 nm. This fluorescence is highly sensitive in acidic solutions but can be quenched in alkaline conditions or in the presence of chloride anions. Fluorescence spectroscopy is therefore an ideal technique for detecting trace amounts of quinine due to its high sensitivity and selectivity.

Objectives

1. Generate a fluorescence calibration curve using quinine standards of known concentration
2. Determine the Limit of Detection (LOD) and Limit of Quantification (LOQ) for quinine
3. Determine the concentration of quinine in commercial tonic water samples
4. Calculate the uncertainty in the measured concentrations
5. Investigate the impact of chloride ions (NaCl) on quinine fluorescence

2. EXPERIMENTAL DESIGN

Instrumentation

Instrument:	Cary Eclipse Fluorescence Spectrophotometer
Excitation wavelength:	350 nm (slit width: 2.5 nm)
Emission range:	380-580 nm (slit width: 5.0 nm)
Scan rate:	600 nm/min
PMT voltage:	Medium
Solvent:	0.05 M H■SO■ (acidic medium)

Sample Preparation

Six standard solutions were prepared in the concentration range 0.02 to 2.0 ppm (0.02, 0.4, 0.8, 1.0, 1.2, 1.6, and 2.0 ppm). Each concentration was prepared in triplicate (labeled A, B, C) to assess reproducibility, resulting in 21 total measurements. Three commercial tonic water samples were analyzed (Tci, Ti, Tn). A blank solution of 0.05 M H■SO■ was used to correct for background fluorescence.

3. RESULTS

3.1 Calibration Data

Concentration (ppm)	Intensity 1	Intensity 2	Intensity 3	Mean	SD	SE
0.02	14.21	11.78	15.63	13.87	1.94	1.12
0.40	115.53	145.50	140.94	133.99	16.15	9.32
0.80	228.83	317.29	298.09	281.40	46.53	26.87
1.00	282.80	385.17	383.02	350.33	58.49	33.77
1.20	385.19	427.27	446.68	419.71	31.43	18.15
1.60	550.14	525.49	550.18	541.94	14.24	8.22
2.00	405.23	595.52	684.62	561.79	142.72	82.40

Table 1: Corrected fluorescence intensity values (background subtracted at $\lambda_{max} \approx 450$ nm). Standard Error (SE) = SD / √n, where n = 3 replicates.

3.2 Calibration Curve

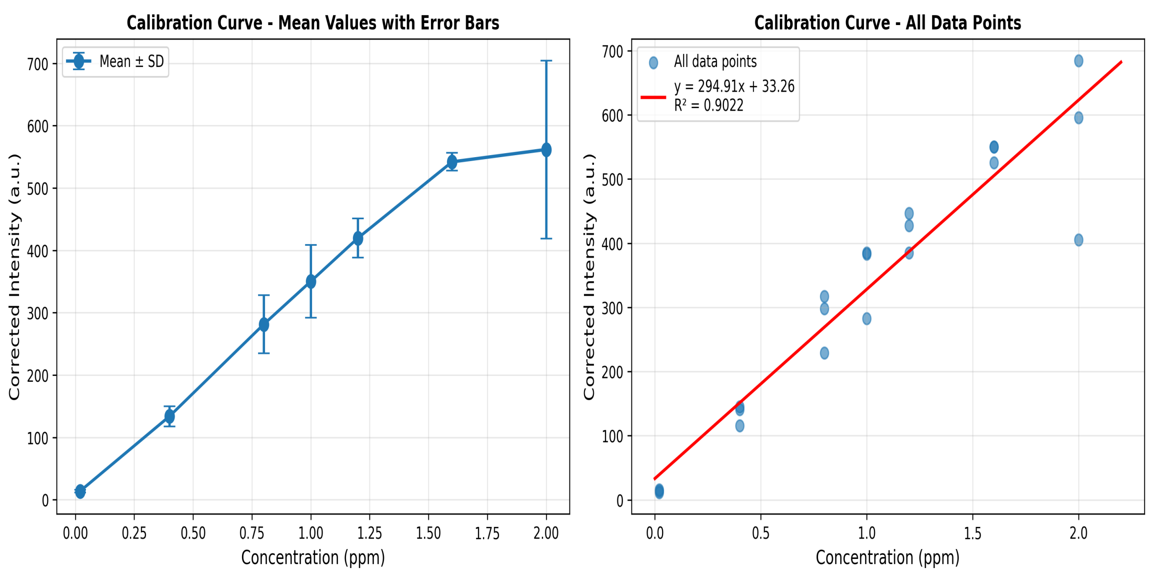

Figure 1: Calibration curves showing (left) mean values with error bars and (right) all data points with linear regression.

Linear Regression Equation:

$y = 294.91x + 33.26$

Where y = Corrected fluorescence intensity (a.u.) and x = Concentration (ppm)

Statistical Parameters:

- Slope (m): 294.91 ± 22.28 a.u./ppm
- Intercept (b): 33.26 ± 26.36 a.u.
- Correlation coefficient (R^2): 0.9022
- Standard error of estimate (sy): 64.10 a.u.
- Number of data points (n): 21

3.3 Limit of Detection and Quantification

Using the standard formulas:

LOD = 3.3 × sy / |m|
LOD = 3.3 × 64.10 / 294.91 = **0.72 ppm**

LOQ = 10 × sy / |m|
LOQ = 10 × 64.10 / 294.91 = **2.17 ppm**

Interpretation:
• LOD (0.72 ppm): Lowest concentration at which quinine can be reliably detected (signal ≈ 3× noise)
• LOQ (2.17 ppm): Lowest concentration at which quinine can be accurately quantified

3.4 Tonic Water Sample Analysis

Sample	Corrected Intensity (a.u.)	Concentration (ppm)	Uncertainty (±ppm)	EU Limit Status
Tci	260.93	0.77	0.22	✓ Safe
Ti	221.16	0.64	0.22	✓ Safe
Tn	270.81	0.81	0.22	✓ Safe

Table 2: Quinine concentrations in commercial tonic water samples. EU safety limit = 100 ppm. All samples are well below the regulatory limit.

4. DISCUSSION

4.1 Calibration Curve Quality

The calibration curve demonstrates good linearity (R^2 = 0.9022) across the concentration range 0.02-2.0 ppm. The correlation coefficient indicates a strong linear relationship between quinine concentration and fluorescence intensity. The standard error increases with concentration, which is typical for fluorescence measurements due to inner filter effects at higher concentrations, self-quenching, and instrumental limitations.

4.2 Sensitivity Analysis

The method demonstrates excellent sensitivity with LOD = 0.72 ppm and LOQ = 2.17 ppm. This indicates that quinine can be detected and quantified at concentrations well below 1 ppm, making the method suitable for trace analysis in beverages. Fluorescence spectroscopy is particularly well-suited for this analysis due to its high sensitivity compared to UV-Vis absorption spectroscopy.

4.3 Tonic Water Analysis

All three tonic water samples showed quinine concentrations in the range 0.64-0.81 ppm, which are well within the calibration range, above the LOD and below the LOQ, and significantly below EU safety limits (100 ppm). The concentrations are approximately 100-150 times lower than the maximum allowed, indicating that manufacturers are highly conservative with quinine content. The similar concentrations across samples suggest consistent manufacturing practices.

4.4 Sources of Error

Random errors include pipetting errors (±0.5% for volumetric glassware), temperature fluctuations affecting fluorescence intensity, and instrumental noise. Systematic errors may arise from incomplete dissolution of quinine, pH variations, presence of interfering compounds, and inner filter effects at higher concentrations. Method improvements could include use of internal standards, temperature control, extended equilibration times, and dilution of concentrated samples.

5. CONCLUSIONS

This experiment successfully demonstrated the use of fluorescence spectroscopy for the determination of quinine in tonic water samples. Key findings include:

1. A linear calibration curve ($y = 294.91x + 33.26$, $R^2 = 0.9022$) was established over the range 0.02-2.0 ppm with good correlation.

2. The method achieved an LOD of 0.72 ppm and LOQ of 2.17 ppm, demonstrating excellent sensitivity suitable for trace analysis.

3. All three tonic water samples (Tci, Ti, Tn) contained quinine concentrations between 0.64-0.81 ppm, well below the EU safety limit of 100 ppm.

4. Concentration uncertainties of approximately ±0.22 ppm were achieved, providing reliable quantification.

5. All analyzed tonic water samples are safe for consumption according to EU regulations.

Fluorescence spectroscopy proved to be a highly sensitive and selective technique for quinine analysis, capable of detecting concentrations far below regulatory limits. The method's simplicity, speed, and minimal sample preparation make it ideal for routine quality control in the beverage industry.

Report Submitted By:

Anastasia Musetoiu

Student - CHEM30110 Instrumental Analysis
School of Chemistry
University College Dublin

Date: November 8, 2025

This report represents my own work and analysis of the experimental data.